

A Hamiltonian-Path Criterion from Bipartite Number and Average Eccentricity

A proof of Written on the Wall II, Conjecture 198a

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Abstract. For a finite nontrivial connected simple graph G , let $b(G)$ be the maximum order of an induced bipartite subgraph and let $\overline{\text{ecc}}(G)$ be the average vertex eccentricity. We prove that

$$b(G) \leq 2 + \overline{\text{ecc}}(G) \implies G \text{ has a Hamiltonian path.}$$

This is the statement recorded as Written on the Wall II, Conjecture 198a. The proof is driven by the tight chain $D + 1 \leq b(G) \leq 2 + \overline{\text{ecc}}(G) \leq D + 2$, where $D = \text{diam}(G)$. In the first equality case, every vertex outside a diametral path belongs to a clique layer that can be inserted into that path. In the second, the graph is self-centered of diameter two, and the Chvátal–Erdős traceability theorem applies.

1. Statement and notation

All graphs in this note are finite and simple. For a connected graph G , write

$$D = \text{diam}(G), \quad r = \text{rad}(G), \quad \overline{\text{ecc}}(G) = \frac{1}{|V(G)|} \sum_{v \in V(G)} \text{ecc}_G(v).$$

The *bipartite number* $b(G)$ is the largest cardinality of a set $S \subseteq V(G)$ for which the induced graph $G[S]$ is bipartite. A graph is *traceable* if it has a Hamiltonian path.

Theorem 1 (Written on the Wall II, Conjecture 198a).

Let G be a nontrivial connected graph. If

$$b(G) \leq 2 + \overline{\text{ecc}}(G),$$

then G is traceable.

We use two classical facts. First,

$$b(G) \geq 2r, \tag{1}$$

as recorded by DeLaViña, Pepper, and Waller [DPW07, Theorem 2]. Second, the Chvátal–Erdős path theorem says that an s -connected graph with no independent set of $s + 2$ vertices is traceable [CE72, Theorem 2].

2. Proof

Choose a diametral geodesic

$$P = v_0 v_1 \cdots v_D.$$

A geodesic is induced, so P is an induced bipartite graph of order $D + 1$. Since every eccentricity is at most D ,

$$D + 1 \leq b(G) \leq 2 + \overline{\text{ecc}}(G) \leq D + 2. \quad (2)$$

Because $b(G)$ is integral, only two cases remain.

Case 1: $b(G) = D + 1$

Fix $x \notin V(P)$. The induced graph on $V(P) \cup \{x\}$ has order $D + 2 > b(G)$, hence is not bipartite. Thus x has neighbors on P of both parities. On the other hand, any two P -neighbors v_i, v_k of x satisfy $|i - k| \leq 2$: otherwise $v_i x v_k$ would shorten the corresponding section of the geodesic P .

Let j be the least index for which $x v_j \in E(G)$. The preceding paragraph gives

$$N_G(x) \cap V(P) \in \{\{v_j, v_{j+1}\}, \{v_j, v_{j+1}, v_{j+2}\}\}, \quad (3)$$

where the second alternative is omitted when $j + 2 > D$. For $0 \leq j < D$, let X_j consist of the vertices outside P whose least neighbor index is j .

Each X_j is a clique. Indeed, if distinct $x, y \in X_j$ were nonadjacent, then

$$(V(P) \setminus \{v_{j+1}\}) \cup \{x, y\}$$

would induce a bipartite graph of order $D + 2$. To see the bipartition directly, retain the alternating coloring of P : the only remaining possible P -neighbors of x or y are v_j and v_{j+2} , which have the same color, and assign both x and y the opposite color. This contradicts $b(G) = D + 1$.

Now order the vertices as

$$v_0, X_0, v_1, X_1, \dots, v_{D-1}, X_{D-1}, v_D. \quad (4)$$

using any order inside each X_j and simply omitting empty layers. Consecutive vertices are adjacent: every X_j is a clique, and every member of X_j is adjacent to both v_j and v_{j+1} . All vertices occur exactly once, so (4) is a Hamiltonian path.

Case 2: $b(G) = D + 2$

Equality throughout the right half of (2) gives $\overline{\text{ecc}}(G) = D$. Every vertex therefore has eccentricity D , so G is self-centered and $r = D$. Combining this with (1) yields

$$2D = 2r \leq b(G) = D + 2,$$

and hence $D \leq 2$. The case $D = 1$ is impossible: a nontrivial diameter-one graph is complete and has bipartite number 2, not $D + 2 = 3$. Consequently,

$$D = r = 2, \quad b(G) = 4. \quad (5)$$

The graph G has no cut vertex. Otherwise, let c be a cut vertex. For any $w \neq c$, choose y in a component of $G - c$ different from the component containing w . Every w - y path passes through c , and a path of length at most two must therefore be wcy . Hence every vertex other than c is adjacent to c , giving $\text{ecc}_G(c) = 1$, contrary to self-centeredness with radius two. Since $D = 2$ also implies $|V(G)| \geq 3$, the graph is 2-connected.

Finally, if I is a maximum independent set, then $I \neq V(G)$ because G is connected and nontrivial. Adding any vertex outside I produces an induced bipartite graph, so $b(G) \geq |I| + 1$. By (5), $\alpha(G) \leq 3$. Hence this 2-connected graph has no independent set of four vertices. The Chvátal–Erdős path theorem, with $s = 2$, supplies a Hamiltonian path. This completes the proof of Theorem 1. $\square$

3. Formalization and reproducibility

The theorem is represented in the Formal Conjectures repository over a finite type α by a connected simple graph G , with $b(G)$ coerced to $\mathbb{R}$ and average eccentricity defined as a real average. The proof artifact accompanying this manuscript is organized around the following independently checkable interfaces:

1. a diametral geodesic and the lower bound $D + 1 \leq b(G)$;
2. the integral dichotomy $b(G) \in \{D + 1, D + 2\}$;
3. the attachment and clique-layer lemmas (3);
4. the self-centered diameter-two reduction; and
5. a direct longest-path traceability endpoint matching the repository statement:

$$\exists ab, \quad \exists p : G.\text{Walk } a \ b, \quad p.\text{IsHamiltonian}.$$

All of these components, including both equality branches and the classical inequality (1), are checked in Lean without proof placeholders. Their top-level composition has the exact hypotheses and conclusion of the repository conjecture.

Artifact status. Lean 4.27 kernel checks pass for the exact repository theorem, including $b(G) \geq 2 \text{ rad}(G)$, with no proof placeholders.

Research provenance and generative-AI use disclosure. I used a near-autonomous, iteratively prompted research workflow to identify and pursue this problem. I began by examining other open problems reported as solved during the preceding several days, comparing their statements and proof strategies to assess which nearby conjectures might be approachable. The resulting model outputs placed this conjecture among comparatively steeper targets, and I selected it for sustained work.

I then used GPT-5.6 Sol in local Codex at extra-high and ultra reasoning settings, cloud-hosted ChatGPT work, and individual ChatGPT 5.6 Pro sessions for iterative research, proof development, Lean formalization, and verification, continuing until the exact theorem passed the Lean kernel. GPT-5.6 Sol at extra-high reasoning was used for repository preparation; Claude Fable 5 and Claude Sonnet 5 were used for limited supplementary searches. Model names in the displayed byline and this disclosure identify software used in the workflow, not independent agency, authorship, or responsibility. I am the sole human author and take responsibility for the claims and submitted material.

Prior-solution search and research status

The statement appears as Conjecture 198a in Written on the Wall II [DeL] and in the Formal Conjectures snapshot [The26]. A targeted search completed on 23 July 2026 covered the current upstream statement, its open and closed issue and pull-request history, public GitHub code indexed under the exact theorem identifier, and exact-phrase web queries. It found copies of the open conjecture and benchmark tasks, but no evidence in the searched sources of a proof or verified formalization predating this work. The search was not an exhaustive literature review, does not establish worldwide novelty or priority, and supports no claim of a first proof.

References

- [CE72] Václav Chvátal and Paul Erdős. A note on hamiltonian circuits. *Discrete Mathematics*, 2(2):111–113, 1972.
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- [The26] The Formal Conjectures Authors. Written on the wall ii, graph conjecture 198a. Formal Conjectures repository, 2026. Snapshot commit e751934294a381afd2d5fc1124c5953c8e25f9fa, accessed 23 July 2026.